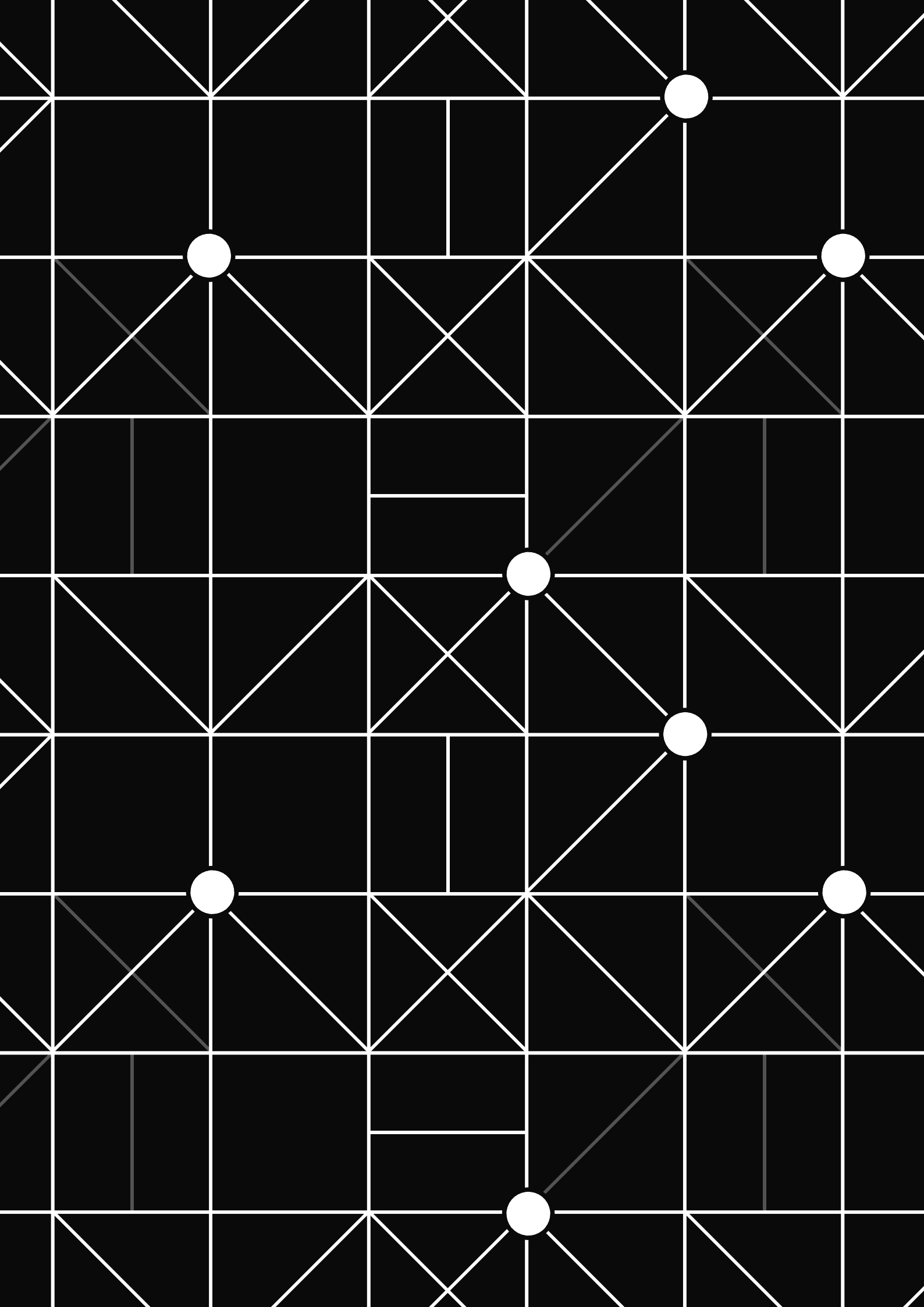




Navigation in a grid-world with obstacles:

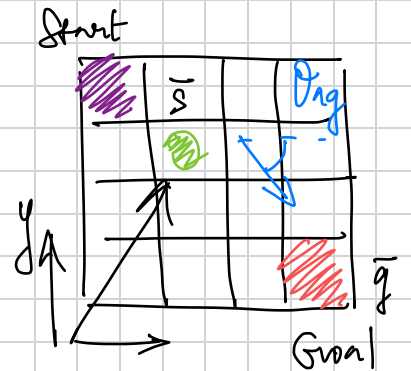
State of agent: $\bar{s} = \{x, y\}$, $\bar{a} = \{N, S, E, W\}$

Goal: $\bar{g}$

→ Geocentric policy, $\pi_G(a)$:

Orientation vector from $\bar{s}$ to goal $\bar{g}$ and the action direction $\bar{a}$:

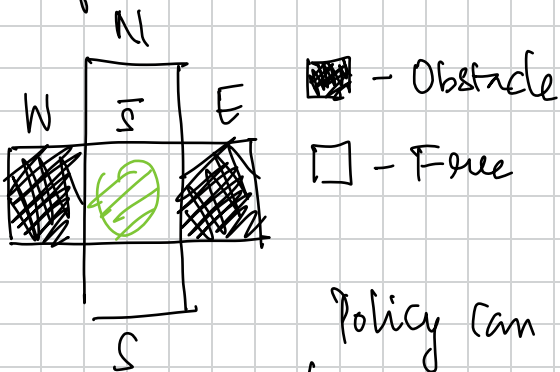
$$\cos \theta_{ag} \equiv \phi_a = \frac{\bar{a} \cdot (\bar{g} - \bar{s})}{|\bar{a}| |\bar{g} - \bar{s}|} = \frac{\hat{e}_a \cdot \hat{h}}{\frac{\bar{a}}{|\bar{a}|} \cdot \frac{\bar{g} - \bar{s}}{|\bar{g} - \bar{s}|}}$$



If $\bar{a} = \hat{e}_a$, the policy^x has high probability of choosing that if it aligns with $\hat{h}$.

→ Egocentric policy, $\pi_E(a)$:

The egocentric policy is a function of the local occupancy of the grids in the neighbourhood:



We can define occupancy of the neighbourhood,

$$N_s = \{O_N, O_S, O_E, O_W\}$$

Policy can take either $\{0, 1\}$ north/south as they are not occupied

0 - Free
1 - Occupied

$$\pi_E(a|s) = \frac{\exp[-\alpha O_a]}{\sum_a \exp[-\alpha O_a]} \text{ where } O_a = \{0, 1\}$$

→ $\alpha > 0$ acts as -ve β . $\sum_a \exp[-\alpha O_a]$ for free obstacle

→ $\pi_E(a|s)$ can take any action away from obstacle.

Let us say there are $k \in \{0, 1, 2, 3, 4\}$ sites blocked, then the partition function becomes,

$$Z_E = k e^{-\alpha} + (4-k) e^0 = (4-k) + k e^{-\alpha}$$

For action along a free site the prob. is $P(a_{\text{free}}) = \frac{(4-k)}{Z_E}$,

and blocked site is, $P(a_{\text{blocked}}) = \frac{k e^{-\alpha}}{Z_E}$.

→ Combined policy, $\pi(a|s) = \gamma \pi_G(a|s) + (1-\gamma) \pi_E(a|s)$
for $\gamma \in [0, 1]$.

Continuum limit of $\pi(a|s)$:

The probability of finding the agent at a particular location $\bar{u}$ at a time instant t can be derived using Master equation:

$$\frac{\partial P(\bar{u}, t)}{\partial t} = \sum_{\bar{a}} [\pi(\bar{a}|\bar{u}-\bar{a}) P(\bar{u}-\bar{a}, t) - \pi(\bar{a}|\bar{u}) P(\bar{u}, t)]$$

After a standard set of steps, it is easy to see that the Fokker-Planck equation can be written as

$$\frac{\partial P(\bar{u}, t)}{\partial t} + \bar{\nabla} \cdot [\bar{M}^1(\bar{u}) P(\bar{u})] = \frac{1}{2} \bar{\nabla} \bar{\nabla} : [\bar{M}^{(2)} P(\bar{u})]$$

n^{th} - moment of the policy $\rightarrow \bar{M}^{(n)}(\bar{u}) = \sum_{\bar{a}} \bar{a}^{\otimes n} \pi(\bar{a}, \bar{u})$.

1st moment of $\bar{\pi}_G(a|s)$:

$$\bar{M}_G^1(\bar{h}) = \sum_{\bar{a}} \bar{a} \bar{\pi}_G(\bar{a}|\bar{h}). \quad \text{Assume an arbitrary action, } \bar{a} = \epsilon \hat{e}_a.$$

$$\bar{\pi}_G(\bar{a}|\bar{s}) = \frac{\exp[\beta \phi_a]}{\sum_{a'} \exp[\beta \phi_{a'}]} \Rightarrow \bar{M}_G^1 = \epsilon \hat{e}_x \bar{\pi}_G(\hat{e}_x) - \epsilon \hat{e}_x \bar{\pi}_G(-\hat{e}_x) + \epsilon \hat{e}_y \bar{\pi}_G(\hat{e}_y) - \epsilon \hat{e}_y \bar{\pi}_G(-\hat{e}_y)$$

$$\phi_{\hat{e}_x} = \hat{e}_x \cdot \hat{u}, \quad \phi_{\hat{e}_y} = \hat{e}_y \cdot \hat{u}$$

$$\phi_{-\hat{e}_x} = -\hat{e}_x \cdot \hat{u}, \quad \phi_{-\hat{e}_y} = -\hat{e}_y \cdot \hat{u}$$

$$\bar{M}_G^1 = \frac{\epsilon \sinh(\beta \phi_{\hat{e}_x})}{\cosh(\beta \phi_{\hat{e}_x}) + \cosh(\beta \phi_{\hat{e}_y})} \hat{e}_x$$

$$+ \frac{\epsilon \sinh(\beta \phi_{\hat{e}_y})}{\cosh(\beta \phi_{\hat{e}_x}) + \cosh(\beta \phi_{\hat{e}_y})} \hat{e}_y$$

$$\bar{Z}_G = \exp(\beta \phi_{\hat{e}_x}) + \exp(-\beta \phi_{\hat{e}_x}) + \exp(\beta \phi_{\hat{e}_y}) + \exp(-\beta \phi_{\hat{e}_y})$$

$$\frac{\bar{Z}_G}{\epsilon} = 2 \cosh(\beta \phi_{\hat{e}_x}) + 2 \cosh(\beta \phi_{\hat{e}_y})$$

1st moment of $\bar{\pi}_E(a|s)$: $\bar{M}_E^1(\bar{h}) = \sum_{\bar{a}} \bar{a} \bar{\pi}_E(\bar{a}|\bar{h})$

$$\bar{Z}_E = (4-k) + k e^{-\alpha} \quad M_{Ex} = \epsilon \cdot \bar{\pi}(\hat{e}_x|\bar{h}) - \epsilon \cdot \bar{\pi}(-\hat{e}_x|\bar{h})$$

Here O_i is the occupancy of the i -th side

$$= \epsilon \frac{e^{-\alpha O_{+x}} - e^{-\alpha O_{-x}}}{\bar{Z}_E} \sum_j e^{-\alpha O_j} \text{ where } j \in \{+x, -x, +y, -y\}$$

$$M_{Ey} = \epsilon \frac{e^{-\alpha O_{+y}} - e^{-\alpha O_{-y}}}{\bar{Z}_E}$$

2nd moment of $\bar{\pi}_E(a|s)$:

$$M_{Exx}^2(\bar{h}) = \epsilon^2 \frac{\bar{\pi}(\hat{e}_x) + \bar{\pi}(-\hat{e}_x)}{\bar{Z}_E} = \epsilon^2 \frac{\bar{Z}_{Ex}}{\bar{Z}_{Ex} + \bar{Z}_{Ey}}$$

$$M_{E_y}^2(\bar{r}) = \epsilon^2 \frac{Z_{E_y}}{Z_{E_x} + Z_{E_y}}$$

Dough

$$D_{+x} = \gamma_d \cdot \partial_x D \cdot \epsilon$$

$$D_{-x} = -\gamma_d \cdot \partial_x D \cdot \epsilon$$

$$\frac{e^{-\alpha D_{+x}} - e^{-\alpha D_{-x}}}{Z_E}$$

Next steps:

→ Derive 2nd moment for $\bar{T}_G(a|S)$

→ Take the continuum limit

by Taylor expanding ϵ

→ Find the non-linear advection and diffusion as a function of $\epsilon, \gamma, \beta, \alpha, \theta_{ag}, \underbrace{D(\bar{r})}_{\text{obstacle density}}$.

obstacle density

$$M_{E_x} = \frac{e^{-\alpha D_{+x}} - e^{-\alpha D_{-x}}}{Z_E} = \frac{[1 - \alpha \gamma_d \cdot \epsilon \cdot \partial_x D] - [1 + \alpha \gamma_d \cdot \epsilon \cdot \partial_x D]}{()} = -\alpha \epsilon \cdot \partial_x D$$

→ Linearize FP eqn. to get LFP [assuming θ_{ag} is small]

→ Calculate first passage prob. $P(g | \bar{v}, D_{\perp}, D_{\parallel}, D(\bar{r}))$